

CSE 4100/ISY4100/ITE4100

Undergraduate Research Project (*Preliminary Proposal*)

Title of Research Project: Adoption and Utilization of e-Government Services in Guyana: Challenges and Opportunities

Student Name: Gaianand Ramnarace

USI: 1045379

Assigned Coach: Seon Renville

Background to the Problem

The Guyanese Government has put out many e-Government services, an example of one of these services is GRAs (Guyana Revenue Authority) 'Padna app, this app was released to the public with the aim of making tax services more accessible for the average Guyanese citizen(Padna – Guyana Revenue Authority, n.d.). Despite this initiative along with others such as the Guyana Civil Aviation Authority portal and the Central Housing & Planning Authority portal which make services offered by these respective agencies accessible to the Guyanese population via the web, the adoption and uptake of e-Government services still remain low in hinterland and remote communities where connectivity and digital readiness are below national averages (UNDP Guyana, 2024; UNDP, 2023).

The most common factors that make it difficult for the adoption and longevity of these kinds of services are: lack of digital literacy, unreliable internet access, UI/UX issues and data privacy concerns among demographics (Méndez-Rivera, 2023; World Bank, n.d.). How people view things such as: usefulness, the ease of use, trust and facilitating conditions are main factors when it comes to adoption. (Davis, 1989; Venkatesh et al., 2003). For small developing nations, the awareness campaigns, institutional capacity and inclusive design are also major factors. (Williams et al., 2015; Jackman et al., 2015).

When it comes to Guyana, there is very limited research to inform policy and design choices, although recent national assessment and project evaluations have started to map out these gaps. (UNDP Guyana, 2024; UNDP, 2023; Saunders-Hastings et al., 2018).Because of this the project aims to identify what are the socio-technical factors

impacting the adoption and continued use of e-Government services in Guyana. A survey instrument using TAM/UTAUT methods will be used to carry out this study (Davis, 1989; Venkatesh et al., 2003). The goal is to translate findings into something tangible for the Government.

The contribution to be made from this study is a roadmap for communication, training and UX improvements to increase uptake and the longevity, especially when it comes to underdeveloped areas (Guyana REDD+ Investment Fund/Office of the Prime Minister & UNDP project evaluations, 2023; Méndez-Rivera, 2023).

Statement of the Problem:

Despite the availability of e-Government services, the adoption in Guyana is held back by trust, usability, access and awareness obstacles that are not yet thoroughly investigated (UN DESA, n.d.; UNDP, 2023). These hindrances are especially prevalent in the remote hinterland areas where infrastructure and facilitating conditions are weaker.(UNDP Guyana, 2024).

Preliminary Review of the Literature:

A systematic review of the literature highlights trust, perceived usefulness, ease of use, and service quality as core predictors of e-Government adoption, while contextual factors (infrastructure, literacy, policy) condition outcomes (Aleisa, 2024; Méndez-Rivera, 2023). Studies in the Caribbean and comparable regions suggest that clear value propositions, simplified interfaces, and targeted outreach improve uptake of e-Government services (Jackman et al., 2015; Rodríguez-Correa et al., 2025).

On top of these insights, this study will use analysis gained from TAM/UTAUT (views of usefulness, views on the ease of use, the expected performance, the effort expected, social influence, facilitating conditions) measure their effects on intention and usage, and analyze the feedback to find design and implications specific to Guyana (Davis, 1989; Venkatesh et al., 2003; Venkatesh et al., 2012).

Rodríguez-Correa et al. (2025) use the TAM method to study e-government adoption among female university students in Medellín, Colombia. They set out to find ease of use and the views on usefulness. Compared with that study, this study on Guyana's population is different both geographically and demographically. The PLOS ONE paper focuses on only female students in an urban area in a Latin American city as opposed to this study which will target the much broader population of Guyana, including both rural and hinterland regions with lower connectivity and different socio-cultural factors which were omitted in the PLOS ONE study, because of these differences, the findings from the PLOS ONE paper would not be applicable to this study. The differences justify the need for local ground research because of Guyana's smaller and more dispersed population along with its inferior infrastructure (Rodríguez-Correa et al., 2025).

Draft Research Question:

Which factors strongly affect the adoption and the continued use of the e-Government services?

(This research question will follow prior work in global studies on e-Government adoption and the factors affecting it; see Venkatesh et al., 2003; Méndez-Rivera, 2023; UNDP, 2023.)

References (APA 7th)

- Aleisa, N. (2024). Key factors influencing the e-government adoption: a systematic literature review. *Deleted Journal*, 1(1), 14–31. <https://doi.org/10.1108/jidt-09-2023-0016>
- Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, 13(3), 319. <https://doi.org/10.2307/249008>
- Guyana REDD+ Investment Fund / Office of the Prime Minister. (2022). ICT and e-services for hinterland, poor and remote communities: Project report. <https://www.guyanareddfund.org/wp-content/uploads/2024/01/ICT-Project-Report-2022.pdf>
- Thomas, T., Singh, L., Gaffar, K., Thakur, D., Jackman, G.A., Thomas, M., Gajraj, R., Allen, C. & Tooma, K. (2014). Measurement invariance of the UTAUT constructs in the Caribbean. *International Journal of Education and Development using ICT*, 10(4),. Open Campus, The University of the West Indies, West Indies. Retrieved October 23, 2025 from <https://www.learntechlib.org/p/150707/>.
- Méndez-Rivera, C., Patiño-Toro, O., Valencia-Arias, A., & Arango-Botero, D. (2023b). Factors Influencing the Adoption of E-Government Services: A Study among University Students. *Economies*, 11(9), 225. <https://doi.org/10.3390/economies11090225>
- padna – Guyana Revenue Authority. (n.d.). <https://www.gra.gov.gy/padna/>
- Rodríguez-Correa, P. A., et al. (2025). Factors influencing the adoption of e-government: Evidence from recent empirical studies. *PLOS ONE*. <https://journals.plos.org>
- Rodríguez-Correa, P. A., Méndez-Rivera, C. A., Patiño-Toro, O. N., Valencia-Arias, A., Conejo, A. L. G. R., León, A. O., Vásquez, J. T. C., & Adan, T. E. A. (2025b). Factors influencing the adoption of e-government by female university students. *PLoS ONE*, 20(1), e0317685. <https://doi.org/10.1371/journal.pone.0317685>
- Saunders-Hastings, P. R., Deonandan, R., & Overhoff, N. (2018c). Likely health impacts of climate change in Guyana: A systematic review. *Journal of Health and Medical Sciences*, 1(1). <https://doi.org/10.31014/aior.1994.01.01.6>
- SmartCitiesWorld. (2022, January 18). Guyana embarks on e-governance project to ensure inclusivity. <https://www.smartcitiesworld.net>
- UNDP. (2023). External mid-term evaluation: ICT Access and E-Services for Hinterland, Poor and Remote Communities. <https://erc.undp.org/evaluation/documents/download/22993>

UNDP Guyana. (2024, November 7). National Digital Readiness Assessment launched. United Nations Development Programme Guyana.
<https://www.undp.org/guyana/press-releases/national-digital-readiness-assessment-launched>

EGOVKB | United Nations > Data > Country Information. (n.d.-b).
<https://publicadministration.un.org/egovkb/en-us/Data/Country-Information/id/72-Guyana>

Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view. *MIS Quarterly*, 27(3), 425–478.

Williams, M. D., Rana, N. P., & Dwivedi, Y. K. (2015). The unified theory of acceptance and use of technology (UTAUT): A literature review. *Journal of Enterprise Information Management*, 28(3), 443–488.

World Bank. (n.d.). E-Government Development Index (EGDI).
https://data360.worldbank.org/en/dataset/UN_EGDI